HW 4 Core: Rotating Polarization with a Chain of Polarizers

Week 4 — Julia script submission

Submission: submit your Julia script on the course Moodle page.

Core homework: a chain of polarizers

We start with light linearly polarized along x, so the electric field vector is $E_0 = (1, 0)$. We send it through $n + 1$ ideal polarizers whose transmission axes step evenly from just above the x-axis up to the y-axis ($90°$): for a given n the axes are at

$$\theta_k = \frac{90° \, k}{n+1}, \qquad k = 1, 2, \ldots, n+1.$$

For $n = 8$ the axes are therefore at $10°, 20°, \ldots, 80°, 90°$. Each polarizer transmits only the component of the field along its axis, i.e. it applies the projection matrix

$$P(\theta) = \begin{pmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{pmatrix},$$

so after one polarizer the field becomes $E \leftarrow P(\theta_k)E$.

Physics box: a polarizer is a projection

An ideal polarizer with axis along the unit vector $u = (\cos\theta, \sin\theta)$ transmits only the component of the field along u: $E_{\text{after}} = (E \cdot u)u$. This is the linear projection $P(\theta)E$ with $P(\theta) = uu^T$, i.e. the 2×2 matrix above. The length of the outgoing field is $|E|\,|\cos(\phi-\theta)|$ when E makes angle ϕ with the x-axis (Malus's law). A chain of many small rotations projects the field step by step; each step keeps a factor $\cos(\Delta\theta)$, so the total amplitude factor is $\cos(\Delta\theta)^n$ with $\Delta\theta = 90°/(n+1)$, which tends to 1 as n grows. In the limit the chain rotates the polarization by $90°$ with no loss. The product of the projections is not itself a rotation matrix (it is rank 1), but on the input $(1, 0)$ it acts like the $90°$ rotation.

Task

Write a Julia program that reads its input from standard input and prints the transmitted intensities. The program:

- reads from the first line an integer k, the number of values of n;
- reads from the second line the k values of n;
- for each n, in the order given, starts from $E = (1, 0)$, loops $j = 1, \ldots, n+1$, computes θ_j in radians as `deg2rad(90*j/(n+1))`, builds $P(\theta_j)$, and updates $E \leftarrow P(\theta_j)E$;
- computes the outgoing amplitude

  $$|E| = \sqrt{E_1^2 + E_2^2},$$

  and the intensity $|E|^2$, the intensity of the radiation transmitted through the whole chain;
- prints the intensities, one value per line, in the same order as the input values of n, each rounded to six decimals with `round(intensity; digits = 6)`.

You may collect the values in a vector and print them in a final loop. Do not define functions, and use only loops and `let` blocks.

Submission

Submit your program in the Week 4 CodeRunner question on the course Moodle page. The checker supplies the input on standard input and compares the printed output line by line, so print nothing else: no input prompts, no headings, and no extra numbers.

Checking

For the input

```
9
1 2 4 8 16 32 64 128 256
```

the program must print exactly nine lines:

```
0.25
0.421875
0.605429
0.759148
0.864721
0.927931
0.962748
0.981054
0.990445
```

Check the smallest case by hand: for $n = 1$ the axes are 45° and 90°, the first polarizer leaves $(1/2, 1/2)$, the second leaves $(0, 1/2)$, so amplitude 0.5 and intensity 0.25. Notice that the amplitude goes to unity: with enough steps the chain rotates the polarization by 90° with almost no loss. Tell your friends how surprised you are.

Optional extension (not for submission)

Plot the amplitude as a function of n on a log scale for n up to 10^4, and estimate how many polarizers are needed to keep at least 99% of the incident amplitude.